

Savitzky-Golay window ablation numerical diagnostics

No new plots are generated by this diagnostic summary.

W=0.0528807640069 s

acceleration-residual dynamics loss=846.8622437038321
selected rotor delay=0.199136850355 s
selected gimbal delay=0.019398895719 s
optimizer status=0 success=False nfev=80 optimality=1.2624563623830909e-05
final gradient inf-norm=1.2624563623830909e-05
gradient tolerance satisfied=False
Jacobian rank=14 cond(nonzero)=430.4615570220279
finite-difference Jacobian max relative error=4.636641855461156e-12
confidence residual sample count=711

W=0.5 s

acceleration-residual dynamics loss=1.093543572579332
selected rotor delay=0.2 s
selected gimbal delay=0.038250608431 s
optimizer status=1 success=True nfev=4 optimality=2.3125099168908037e-08
final gradient inf-norm=2.3125099168908037e-08
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=409.998268653704
finite-difference Jacobian max relative error=9.175095543339716e-12
confidence residual sample count=658

W=1 s

acceleration-residual dynamics loss=1.0792047277442087
selected rotor delay=0.2 s
selected gimbal delay=0.040217049712 s
optimizer status=1 success=True nfev=4 optimality=1.8594785267112002e-08
final gradient inf-norm=1.8594785267112002e-08
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=406.9245366549322
finite-difference Jacobian max relative error=7.904933618886317e-12
confidence residual sample count=598

W=1.5 s

acceleration-residual dynamics loss=1.1428722572575483
selected rotor delay=0.190958180968 s
selected gimbal delay=0.037856599215 s
optimizer status=1 success=True nfev=4 optimality=2.7666576585261282e-08
final gradient inf-norm=2.7666576585261282e-08
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=410.1436198608981
finite-difference Jacobian max relative error=9.027424324664627e-12
confidence residual sample count=538

W=2 s

acceleration-residual dynamics loss=1.151014936433081
selected rotor delay=0.2 s
selected gimbal delay=0.030154944961 s
optimizer status=1 success=True nfev=4 optimality=7.994731454630255e-08
final gradient inf-norm=7.994731454630255e-08
gradient tolerance satisfied=True

Jacobian rank=14 cond(nonzero)=392.2063177156247
finite-difference Jacobian max relative error=8.60942559528201e-12
confidence residual sample count=478